

# File permissions in Linux

## Project description

The research team at my organization needs to update the file permissions for certain files and directories within the `projects` directory. The permissions do not currently reflect the level of authorization that should be given. Checking and updating these permissions will help keep their system secure. To complete this task, I performed the following tasks:

## Current file permissions

I used the `ls -la` command to list the contents with permissions.

```
researcher2@7448d31460a7:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  3 12:53 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  3 13:47 ..
-rw--w---- 1 researcher2 research_team  46 Mar  3 12:53 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar  3 12:53 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Mar  3 12:53 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Mar  3 12:53 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  3 12:53 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  3 12:53 project_t.txt
```

## Permission String description (drwxrwxrwx):

The first set of permissions concerns the user (r for read, w for write, and x for execute). The second set is for the group. And the third set is for every other user on the system.

Changes in file permissions:

The organization does not allow others to have write access to any files.

I ran this command to comply with the company policy.

```
researcher2@7448d31460a7:~/projects$ chmod o=r project_k.txt
```

It does not have any standard output. This command changes the file permission to read-only for the others.

The organization allows read only on the archived projects, therefore I ran this command.

```
researcher2@7448d31460a7:~/projects$ chmod u=r,g=r .project_x.txt
```

It does not have any standard output. This command changes the file permission to read-only for the user and the group.

## Changes in directory permissions:

The files and directories in the projects directory belong to the researcher2 user.  
Only researcher2 should be allowed to access the drafts directory and its contents.  
I ran this command to comply with this policy.

```
researcher2@7448d31460a7:~/projects$ chmod g= drafts/
```

It does not have any standard output. This command takes away all permissions from the group.

## File permissions after changes:

```
researcher2@7448d31460a7:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  3 12:53 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  3 13:47 ..
-r--r----- 1 researcher2 research_team  46 Mar  3 12:53 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Mar  3 12:53 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Mar  3 12:53 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Mar  3 12:53 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  3 12:53 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Mar  3 12:53 project_t.txt
```

## Summary:

I changed the files and directories permissions to comply with company policies. I ran a total of 3 changes to be more secure and robust. The first step was checking the permissions running `ls -la`, to display all files and directories with permissions. I identified the problems according to the compliance and changed permissions accordingly.